MIL-P-16077C(YD)
12 February 1985
SUPERSEDING
MIL-P-16077B(YD)
30 October 1979

MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, WATER CIRCULATING, ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric-motor-driven, centrifugal, water circulating pumps for use in hot water heating systems and service water distribution systems. The pump units have in-line connections to permit unsupported installation of the units in a straight run of piping.

1.2 Classification. Pumps shall be of the following types, classes, sizes, and intended service, as specified (see 6.2):

Type I - Pump shaft in horizontal plane.
Type II - Pump shaft in vertical plane.

Class 1 - Standard head-capacity (see figure 1).
Class 2 - High head-capacity (see figure 2).

Service A - Hot water heating systems.
Service B - Service water distribution systems.

Sizes - 3/4, 1, 1 1/4, 1 1/2, 2, 2 1/2, and 3 inches.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.1 - Cast Iron Pipe Flanges and Flanged Fittings.
B16.24 - Bronze Pipe Flanges and Flanged Fittings.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump units covered by this specification shall be the in-line, centrifugal, electric-motor-driven type. The motor shall be supported in a mounting bracket attached to the pump casing to form an integrated, self-contained unit. The size of the pump is determined by the size of the discharge.

3.2 First article. When specified (see 6.2), the contractor shall furnish a pump of the type, class, service, and size required for first article inspection and approval (see 4.2.1 and 6.5).

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3.3 Design. The pump unit design shall permit accessibility for maintenance and service in the field, and shall be such as to prevent conditions which may be hazardous to operating personnel. The pump units shall be capable of handling water at temperatures up to 225 degrees Fahrenheit (°F) and shall be furnished complete in all details, including those not specifically mentioned herein.

* 3.4 Standard commercial product. The pumps shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the pumps being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7 Performance. The pumps shall meet or exceed the minimum performance requirements of the applicable head-capacity curve(s) shown in figures 1 and 2. Performance shall be based on pumping water at a temperature of 80°F while a net positive suction head of not more than 20 feet of water is maintained.

3.8 Construction. The pump units shall include a motor, motor mounting bracket, and pump with inlet and outlet connections on opposite sides of the casing on a common centerline. Pump units shall be constructed to permit unsupported installation in a straight run of horizontal or vertical piping. For type I, the horizontal pump shall be connected to the motor by a shaft with a flexible coupling designed to minimize the transmission of motor vibrations to the pump casing and connected piping. For type II, the vertical pump shall be connected to the motor by a thrust-carrying solid extended motor shaft.

3.8.1 Pump casing. Unless otherwise specified (see 6.2), the pump casing and flange shall be made of close-grained cast iron for service A and brass or bronze for service B.

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* 3.8.2 Impeller. Unless otherwise specified (see 6.2), the impeller shall be made of a suitable nonferrous material. When specified (see 6.2), the impeller shall be made of brass or bronze. The impeller shall be open, semi-enclosed, or enclosed type to meet the required performance. The impeller shall be mechanically balanced, and shall be keyed and locked or otherwise securely fastened to the pump shaft to prevent backing-off or loosening with change of rotation.

* 3.8.3 Pump shaft. Unless otherwise specified (see 6.2), the pump shaft shall be alloy steel or nickel alloy. The pump shall have oil lubricated sleeve bearings or grease-lubricated ball bearings. Sleeve bearings shall be lubricated by oil supplied either by a capillary material from an oil reservoir or centrifugally by a slinger affixed to the shaft rotating inside an oil sump chamber.

* 3.8.4 Connections. Direction of flow shall be indicated by permanent markings on the pump casing. Inlet and outlet connection shall be flanged and shall be furnished with threaded companion flanges. Flanges for sizes up to and including 2 inches shall be not less than two-bolt; flanges for sizes over 2 inches shall be not less than four-bolt. Flanges for sizes 2 1/2 inches and over shall conform to ANSI B16.1 and B16.24, as applicable.

* 3.8.5 Seal. A replaceable seal shall be provided between the pump chamber and the bearing housing to prevent leakage of water from the pump chamber to the bearing housing. Seals furnished shall be suitable for use at the temperatures and pressures specified herein.

* 3.8.6 Motor. The motor shall be constructed and rated in accordance with the applicable provisions of NEMA MG 1. Unless otherwise specified (see 6.2 and 6.4), motors supporting loads in excess of 1/2-horsepower (hp) shall be a three-phase, continuous duty, 60 Hertz, 208-230/460 volt (V), with a drip-proof enclosure design, and operate in an ambient temperature of 40° Celsius. For loads of 1/2-hp and less, single-phase motors having a nominal 115V rating may be used. The motor shall be furnished with built-in thermal overload protection and shall have a rated speed not to exceed 1,800 revolutions per minute. The hp rating of the motor, including standard NEMA service factors shall be not less than the maximum brake hp requirement of the pump under any condition of head and capacity, from the manufacturer's maximum rating head to free delivery. Vibration-damping mountings shall support the motor in the mounting bracket.

3.9 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2), pumps shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice.

* 3.10 Identification marking. Identification shall be permanently and legibly marked directly on the pump or on a corrosion-resisting metal plate securely attached to the pump at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

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3.11 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.12 Workmanship.

3.12.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.12.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.12.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.12.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

* 3.12.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. The first article inspection shall be performed on one pump when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4.1, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each pump shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.4 Tests. The first article shall receive the tests of 4.4.1. Each production unit shall receive the test of 4.4.2. Failure to pass any test shall constitute cause for rejection.

4.4.1 Performance test. The pumps shall be tested to determine conformance to the requirements of 3.7. A characteristic head-capacity curve shall be plotted over the entire operating range of the pump while a net positive suction head of not more than 20 feet of water is maintained. During this test run, the electrical input to the motor shall be checked and recorded at increments of head not greater than 1 foot. At no point on the head-capacity curve shall the current demand exceed the ampere rating and NEMA service factor of the motor. Water at temperatures less than 80°F may be used as the test medium, in which case appropriate correction factors shall be applied to the capacities given in figures 1 and 2.

4.4.2 Operational test. The unit shall be connected to a water supply and shall be operated a sufficient period of time to demonstrate that the unit is functioning as specified herein. After completion of the test, the unit shall be drained and serviced for shipment.

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* 4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-P-10603. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

* 5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-10603 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The pump units are intended for use as circulators in forced hot water heating systems and in service water distribution systems where the water temperature for either application does not exceed 225°F.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, service, and size required (see 1.2).
c. Issue of document required if other than as specified (see 2.1.1).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.5).
e. When pump casing and flange are to be other than as specified (see 3.8.1).
f. When material for the impeller is to be other than as specified (see 3.8.2).
g. When the impeller is to be made of brass or bronze (see 3.8.2).
h. When the pump shaft is to be other than as specified (see 3.8.3).
i. When the motor is to be other than as specified (see 3.8.6 and 6.4).
j. When cleaning, treatment, and painting are to be other than as specified (see 3.9).
k. When treatment for fungus resistance is required (see 3.11).
l. When a preproduction pack inspection is required and the time frame required for submission (see 4.2.1 and 4.5).
m. Level of preservation and level of packing required (see 5.1).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 Electrical supply. If three-phase power is not available, single-phase motors may be specified. However, the motor should be rated for the higher systems phase voltage available (see 3.8.6).

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* 6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one pump. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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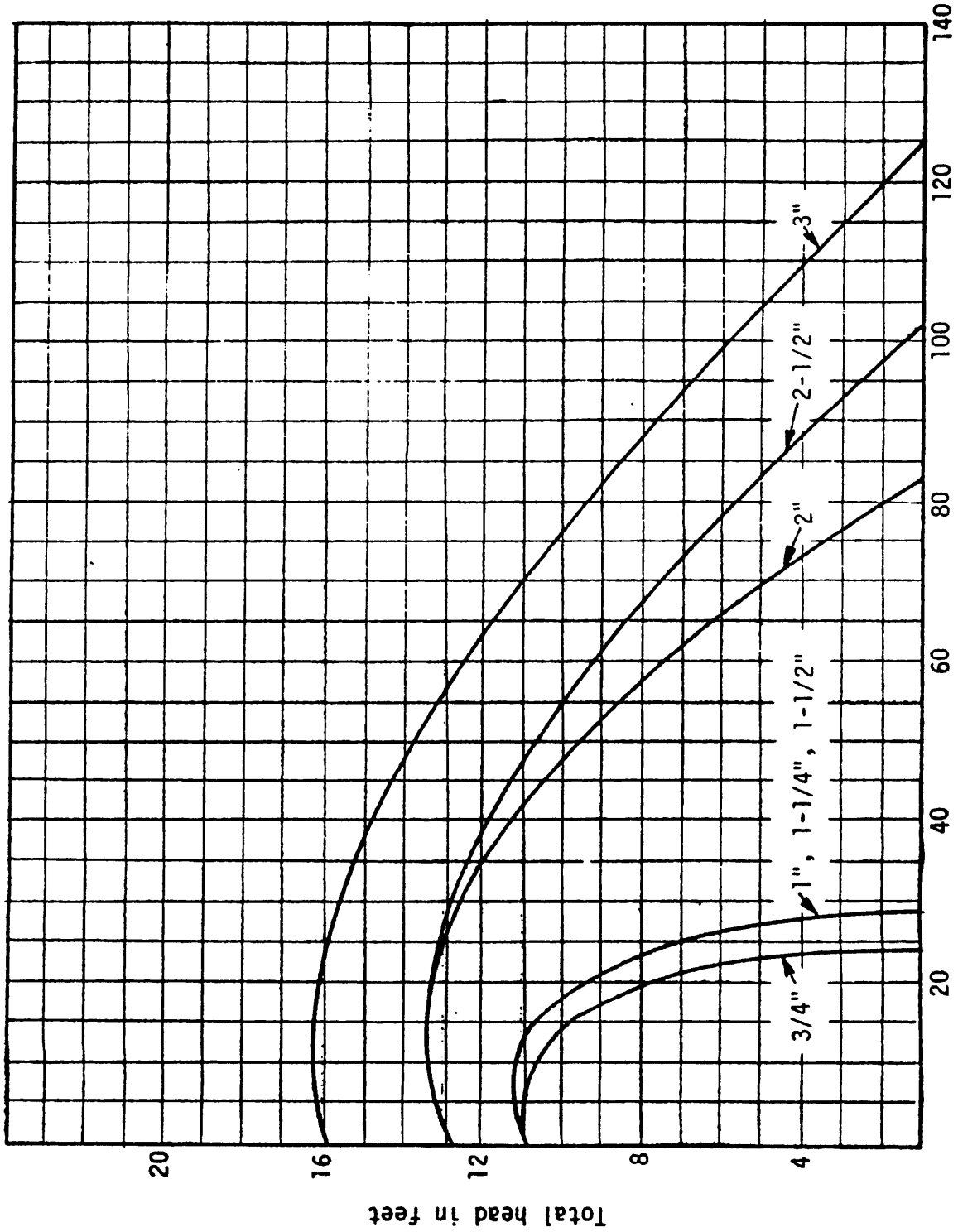


FIGURE 1. Minimum performance requirements for class 1 pumps.

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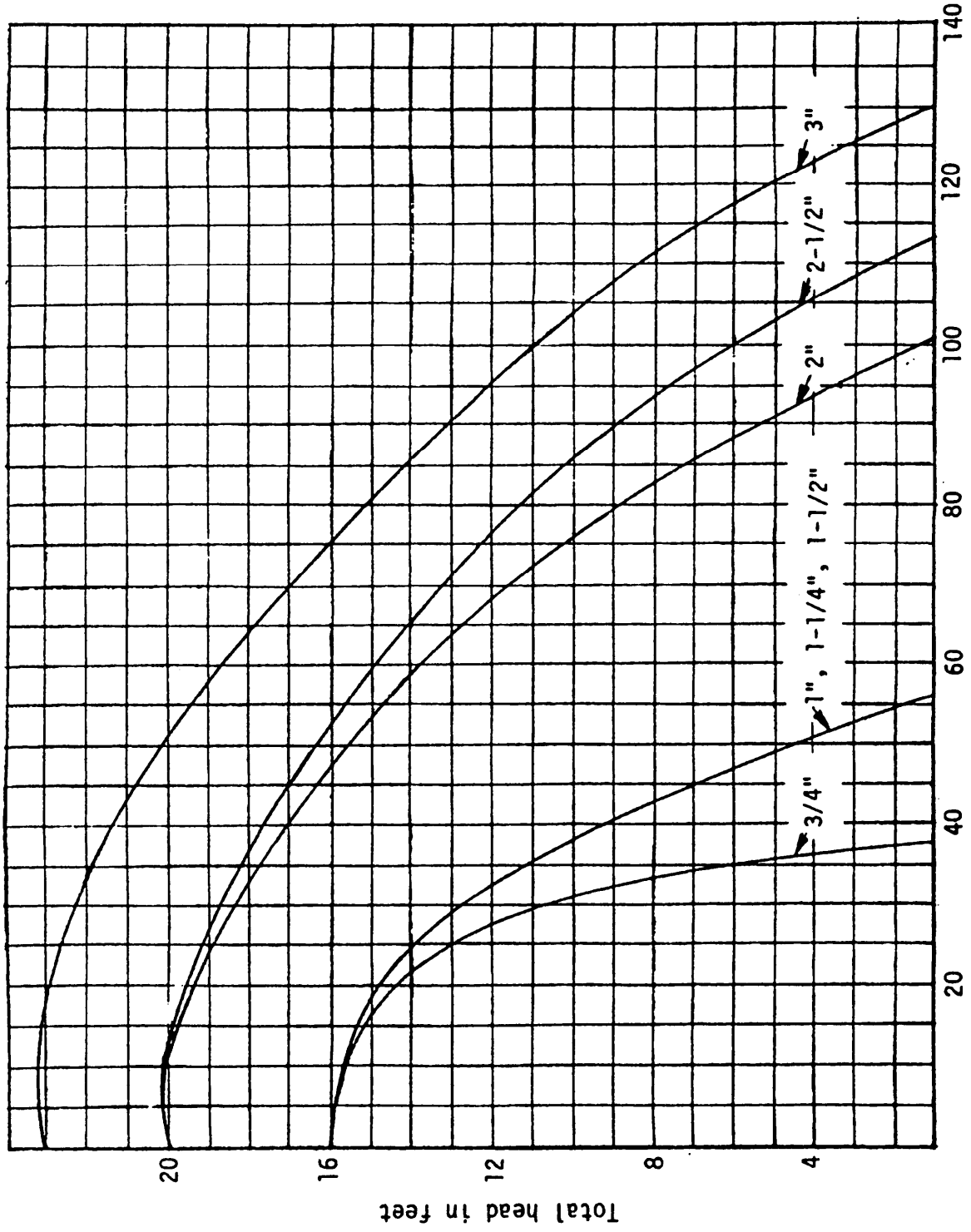


FIGURE 2. Minimum performance requirements for class 2 pumps.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-16077C(YD)	PUMPS, CENTRIFUGAL, WATER CIRCULATING, ELECTRIC-MOTOR-DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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